number of these attention maps, you'll notice that the model tends to pay most attention to the inflection and crossover points of the digit. For the digit 8, it can successfully classify this image as the number 8 just by paying attention to the center of the 8 and the bottom part. You can also notice that in none of the examples is attention given to the added spots of noise in the input; attention is only given to the real digit part of the image, demonstrating that the model is learning to separate the signal from the noise to a large degree.

10.4. Multi-head attention and relational DQN

We've demonstrated that our relational model performs well on the simple task of classifying MNIST digits and furthermore that by visualizing the learned attention maps we can get a sense of what data the model is using to make its decisions. If the trained model keeps misclassifying a particular image, we can inspect its attention map and see if perhaps it is getting distracted by some noise.

One problem with the self-attention mechanism we've employed so far is that it severely constrains the amount of data that can be transmitted due to the softmax. If the input had hundreds or thousands of nodes, the model would only be able to put attention weight on a very small subset of those, and it may not be enough. We want to be able to bias the model toward learning relationships, which the softmax helps promote, but we don't want to necessarily limit the amount of data that can pass through the self-attention layer.

In effect, we need a way to increase the bandwidth of the self-attention layer without fundamentally altering its behavior. To address this issue, we'll allow our model to have multiple attention *heads*, meaning that the model learns multiple attention maps that operate independently and are later combined (figure 10.18). One attention head might focus on a particular region or features of the input, whereas another head would focus elsewhere. This way we can increase the bandwidth through the attention layer but we can still keep the interpretability and relational learning intact. In fact, multi-head attention can improve interpretability because within